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Synchronization of High Speed OFDM links using Variations of Schmidl Cox for SNR Improvement

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***Abstract*—Owing to the growing demand for the high-speed wireless communications which uses Orthogonal Frequency Division Multiplexing (OFDM), there is a need to provide high-quality to overcome various channel impairments appearing over the transmission link. One of the main problems in providing such mechanisms is Doppler shift. which reduces the spectral efficiency of the system, raising a need to determine its value with high accuracy. In order to estimate this frequency shift, first the timing recovery is required. Over the years, Schmidl Cox algorithm has been used to determine this Doppler shift with considerate amount of success. But, with the increasing speed of the vehicle and the worsening channel conditions, this algorithm succumb to failure. The paper discusses the limitations of basic Schmidl Cox algorithm and the necessary improvements to be made, so as to push them to work better at harder conditions too.**

***Index Terms*—High Speed OFDM, Doppler Shift, Timing Synchronization, Frequency Synchronization, Schmidl Cox, Signal to noise Ratio (SNR).**

I. INTRODUCTION

IN any communication system, where there is a burst transmission, it is of utmost importance to maintain timing synchronization, since it serves as the timing reference to the data symbols that follow. In OFDM system this means figuring out the end of the preamble, which in turn gives an estimate of where the data starts. The frequency shift problem is entangled with this timing synchronization, without which the estimation of its shift is not possible. So the frequency estimation and timing synchronization algorithms go hand-in-hand. Several timing recovery algorithms have been proposed in[1],[2] to obtain $'coarse'$ and $'fine'$ timing synchronizations. But the severe effects of the time slip enforces these techniques to be performed regularly at several instances, in order to correct the error being accumulated gradually over time. The preambles reduce bandwidth efficiency considerably, so the algorithms are to be made robust and efficient so as to optimize the number of preambles used.

In Timing Synchronization, the primary objective is to determine where to start the FFT (Fast Fourier Transform)-window. The window need not coincide exactly with the signal position. Due to the FFT cyclic-properties, it can start from any sample within the cyclic-prefix region. This would result in a cyclically-shifted signal. The effects of Timing offset are shown in figure 1. If the window spans samples from two different symbols, it would result in an Inter-Symbol-Interference (ISI).

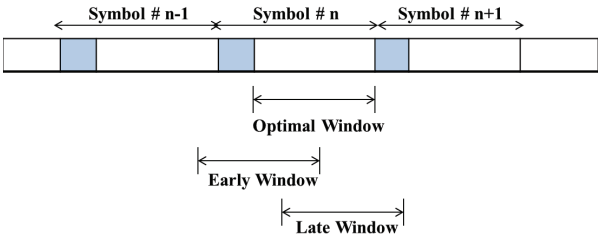


Fig. 1: Effects of Timing Offset

In Frequency Synchronization, the primary objective is to determine the frequency offset present in the signal. Total Frequency offset is the sum of Fractional frequency offset and Integer frequency offset. Effects of frequency offset are shown in figure 2. Due to Fractional frequency offset subcarriers loose orthogonality between them which results in Bit Error Ratio (BER) degradation. Due to Integer frequency offset subcarriers will remain orthogonal but the received data will have cyclically shifted frequency positions compared to the transmitted data. The Schmidl-Cox algorithm[3] proved to

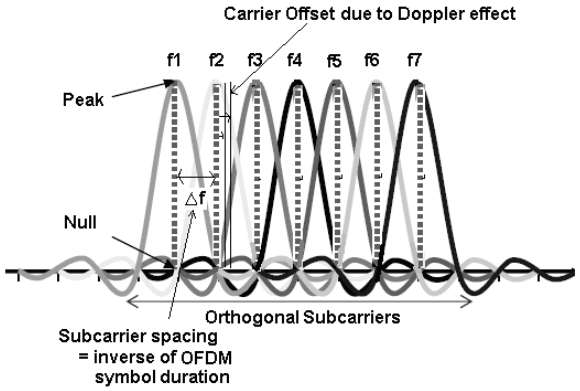


Fig. 2: Effects of frequency Offset

be quite successful even at channel noises of around 6dB SNR and upto 0.01 relative frequency shift. But the Present wireless systems can have worse Doppler shifts as high as 0.7 (relative) and low SNRs. The Schmidl-Cox algorithm alone is

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not capable of handling such harsh conditions. But then there was a scope to improvise upon at several fronts.

But the main problem with respect to this Schmidl-Cox algorithm was the lack of a more specific designation for the timing synchronization in the correlation graph. Determining the end point of the plateau becomes an illusionary task, once the noise factor came into picture. Also, at high levels of noise, the correlations are not capable of carrying out adequate averaging of noise.

In the frequency estimation algorithm, there are two steps. One, to determine the fractional frequency offset and the other, integer offset. There is a standard algorithm by Schmidl-Cox [3] to evaluate each of these offset components using two preamble symbols. The scheme used to find integer offset is very efficient. But the fractional part has significant error at high noise levels. This can be attributed to improper averaging out of noise in the correlation made.

The Paper is organized into 4 sections. In Section II, the basic Schmidl Cox and improvements over it are discussed. In Section III,the proposed techniques are compared with conventional Schmidl Cox using Matlab are presented. Section IV, mentions the summary with concluding remarks.

II. SYNCHRONIZATION ALGORITHM

A. Timing Synchronization Algorithm

1) Schmidl Cox Algorithm: This scheme [3], [4] uses Preamble sequences at regular intervals of time to identify the start of the frame and then to correct it in case of any slip. In each preamble, two (or more) repetitions of the same samples are transmitted, so these parts still remain identical on the receiver side, except for a phase being added because of the frequency shift. So correlation of the signal with a $\frac{OFDMsymbollength}{Numberofrepititions}$ shifted version of itself, would give a maximum when the correlation window is across the two repetitions. *These* repetitions are achieved by transmitting a

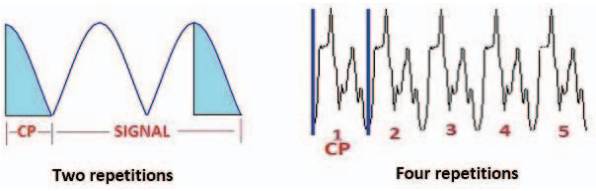


Fig. 3: OFDM Symbol with Preamble Repetitions

pseudo-random (PN) sequence with required number of zeroes (viz., 3 zeroes for four repetitions) between two consecutive symbols. These PN symbols can be chosen randomly from a QPSK constellation, which would enable uniform avg. power over the frequency spectrum. For a 256-point FFT OFDM system, prefer a preamble with four repetitions. Here the correlation is carried out across samples which are spaced N/4 distance apart over 3N/4 samples. So, when the conjugate of the symbol is multiplied with N/4 delayed version of itself, maximum is achieved as a plateau when the correlation interval spans the lobes 1,2,3,4 until its end when it is on 2,3,4,5 as shown in 3. As per Schmidl Cox algorithm, correlation P1(d) is given by

$$Correlation : P1(d) = \sum_{n=0}^{\frac{3*N}{4}-1} r_{d-n} r^*_{d-n-\frac{N}{4}} \quad (1)$$

where r is the recieved ofdm signal, N is OFDM symbol length, d is a variable with values form 0 to 255. As per the definition, plateau occurs at the position of the last lobe in the symbol, i.e., it spans a length of $N/4$ before and until the start of the Cyclic Prefix (CP) of symbol next to preamble.

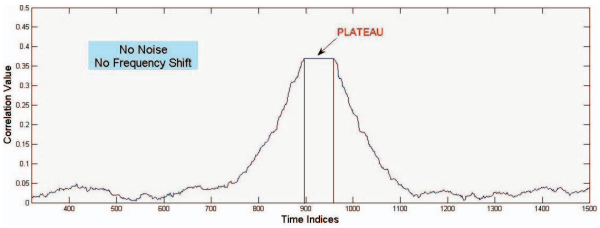


Fig. 4: Correlation graph of Schmidl-Cox algorithm applied on an ideal system with no noise and no frequency shift

But a fine plateau can be observed as shown in figure 4 only in the case of a system with no noise. In the case of frequency shift, although there is phase difference between the samples correlated, its absolute value will still remain the same, keeping the shape of plateau intact. Here 90% estimate procedure can still be carried out to find the points that atleast lie on the plateau. As can be seen in 5, it gets more difficult

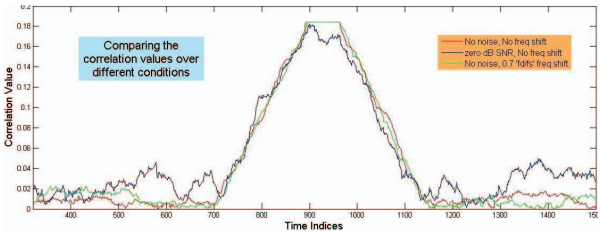


Fig. 5: Correlation graph of Schmidl-Cox algorithm comparing at various conditions

to determine the peak in the graph, making it impossible to find the edge of the plateau.

2) Multiple Correlations: It can be observed that the correlation need not just be confined to the samples that are just N/4 distance away. Even the samples N/2 and 3N/4 distance apart are identical. So, correlation is extended accordingly. Multiple Correlations (P) is described as

$$MultipleCorrelations : P(d) = |\sum_{n=0}^{\frac{3*N}{4}-1} r_{d-n} r^*_{d-n-\frac{N}{4}}| + |\sum_{n=0}^{\frac{2*N}{4}-1} r_{d-n} r^*_{d-n-\frac{2*N}{4}}| + |\sum_{n=0}^{\frac{N}{4}-1} r_{d-n} r^*_{d-n-\frac{3*N}{4}}| \quad (2)$$

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The absolute values of the correlations that are executed for samples spaced N/4, N/2, 3N/4 apart are added, and this cumulative value projects an improved performance than the earlier one. This improvement is attributed to more averaging out of the noise that is done by increasing the correlation window, because of which noise effects have been nullified to some extent. As depicted in figure 6, the shape of plateau is

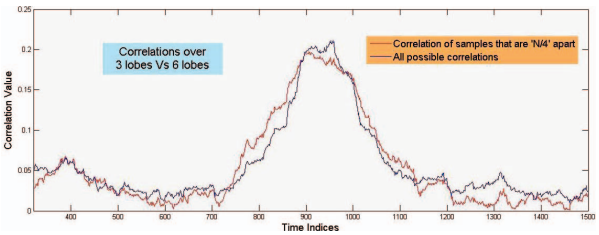


Fig. 6: Comparison of new scheme vs old scheme

more imprinted which makes it all the more easy to pinpoint some point on the plateau, thus increasing the efficiency of the algorithm. The standard method of 90% algorithm can be applied here.

But this would provide efficient results only in the absence of delay spread. The very purpose of including cyclic-prefix is to compensate for this delay-spread, in the presence of which the starting portion of the CP is corrupted. To be more specific, the delay caused when the symbol is passed through the channel would extend into the cyclic prefix of the next symbol. This would effectively reduce the length of the plateau, since the starting few samples in a symbol will no longer be identical to that of the actual signal samples.So if the timing recovered lies in this starting portion of the Cyclic Preefix(CP), certain erroneous samples will be collected in every symbol, which in turn degrades the system performance.

3) Double Schmidl Cox: Although, the above method is preferred to get the timing synchronization to fall in the cyclic prefix, like discussed, due to the constraints of channel, it is not an ideal method. Since the goal is not pinpointed (any point on the plateau) and the noise being not averaged out properly, there is a need for more robust method [4] through which the edge of the symbol is more visible and easy to figure out. On observing the plateau carefully, one can see that it is similar to the preamble with several repetitions, i.e., all the samples are equal (under ideal conditions).

Since the length of the plateau is already small, it can be considered as a symbol with just two identical halves. So upon auto-correlation of these correlated values with $\frac{Ncp}{2}$ (where Ncp is the length of Cyclic Prefix) delayed version of itself, this would give a unique peak, when half the plateau is multiplied by conjugate of the other half of the plateau, i.e., at the end of the symbol. A triangle shaped graph is formed as shown in 7, in which finding the peak must be fairly easier than in the case of a plateau where the peak might not even lie in this region if the noise added to the samples is strong enough. The Double Schmidl cox correlation (S) is given by

$$DoubleSchimdlCox : S(d) = \sum_{n=0}^{\frac{Ncp}{2}-1} P_{d-n} P^*_{d-n-\frac{Ncp}{2}} \quad (3)$$

where d varies from 0 to $\frac{Ncp}{2}$-1.

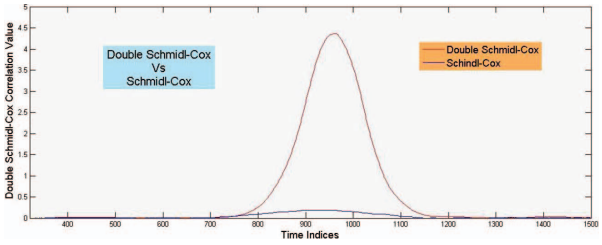


Fig. 7: Comparing the Double Schmidl-Cox to normal Schmidl-Cox method

In the Schmidl-Cox correlation, like discussed, there can be a peak before or after the plateau depending on the error on the samples there respectively. A more robust method is needed as shown in 8. First the peak of the graph must be found out and lets say its amplitude is max. Now, the sample to the left of the peak which has an amplitude (xmax) and a sample to the right of the peak with amplitude (ymax) are to be marked. After Matlab simulations, considerable x is 0.25 and y is 0.15. The average of these two points should ideally give end of the symbol timing. But the error can be positive or negative, which would lead in detecting the timing either before the end of the symbol which is still reasonable or into the next symbol which is not preferred. So the final value is shifted to the left by some distance parameter z which is $\frac{Ncp}{2}$.

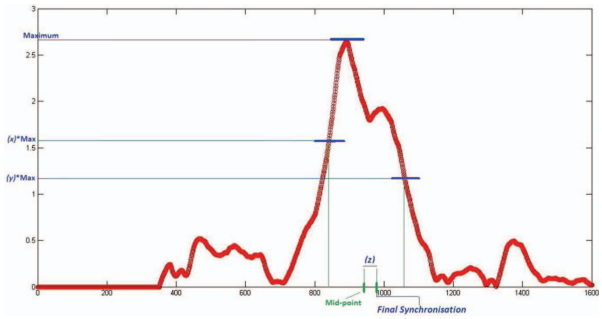


Fig. 8: Mechanism to find the timing synchronization from the Double Schmidl Cox graph

B. Frequency Synchronization Algorithm

1) Schmidl Cox algorithm: Once the timing is realized, the frequency shift in the system must be estimated. The preamble with multiple repetitions again comes into use for this purpose. When the lobes/signals that are identical are passed and collected at the receiver, they remain identical

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except that there will be a phase shift. So the phase of the correlation values on the plateau must give this phase difference. If the conjugate of a sample from the one quarter (half) is multiplied by the sample in another quarter (half), the product will have a phase that is directly proportional to the frequency shift [3]. The phase difference is given by

$$PahseDifference : \phi = 2 * \pi * \Delta f * \frac{T}{4} \tag{4}$$

where Δf is the total frequency Shift, T is the OFDM symbol period. This phase difference will be of the form (2uπ + α), where α is the fractional part and u the integer part. α can be determined directly i.e., it is just the argument of ϕ which follows the condition -π < ϕ < π. The correlation can be carried out across three repetitions, so as to average out more noise and to better estimate the fractional part.

$$P = \sum_{n=0}^{\frac{3*N}{4}-1} r_{m-n} r^*_{m-n-\frac{N}{4}}, \alpha = \arg(P) \tag{5}$$

where m is the timing synchronization output, r is the received signal.

The symbol received can be corrected with the fractional part of frequency offset, where $\bar{\Delta f} = \frac{2*\alpha}{\pi * T}$. The received samples are thus to be multiplied by $\exp(-j2\pi\bar{\Delta f}t)$ for the correction. Now the integer part can be estimated using two preambles. The first preamble has four repetitions, i.e., it has a PN sequence of QPSK symbols at every fourth sample in the frequency domain. The second symbol has a PN sequence of QPSK symbols at all samples in frequency domain, but with half the amplitude, so that both the preamble symbols have the same average power.

Let x_1 be the PN (Pseudo Noise) sequence of 4p+1 sample frequencies of first preamble.Let x_2 be the PN sequence of 4p+1 sample frequencies of second preamble. Now let v be the relative PN sequence, where $v(k) = \frac{x1(k)}{x2(k)}$. Integer part of phase difference implies that the sub-carriers are still orthogonal to each other but the frequency positions itself are shifted. So, at the receiver end, the frequency spectrum shows cyclically versions of the original PN sequences. This shift is to be measured. Let y_1 be the 4p+1 sample frequencies FFT of the corrected first preamble Let y_2 be the 4p+1 sample frequencies FFT of the corrected second preamble The sequence v appears at the output but shifted by 4g positions (where g varies from 0 to Ncp-1) because of the uncompensated frequency shift. So the expression

$$B(g) = \sum_{k=0}^{\frac{N}{4}} y1(k + 4 * g)v^*(k)y2^*(k + 4 * g) \tag{6}$$

This attains the peak value when the value g coincides with the actual integer shift of the system. The total frequency shift is just the sum of the fractional and integer part.

$$Total frequency Shift : \Delta f = \bar{\Delta} f + \frac{4 * g}{T} \tag{7}$$

This method is very efficient, the integer part gives correct results always even at negative SNR values of noise. But the fractional part has significant error because of the noise not being averaged out properly. For a 256-point FFT, maximum fractional offset is 4/256=0.015625.

2) Multiple Correlations: To increase the efficiency, the multiple correlations in the preamble can be exploited for phase (α) estimation. The other possible correlation combinations are calculated and their phases are taken into consideration.

$$\begin{aligned} P1 &= \sum_{n=0}^{\frac{2*N}{4}-1} r_{m-n} r^*_{m-n-\frac{2*N}{4}} \\ P2 &= \sum_{n=0}^{\frac{N}{4}-1} r_{m-n} r^*_{m-n-\frac{3*N}{4}} \end{aligned} \tag{8}$$

Let us consider $p = \arg(P)$, $p1 = \arg(P1)$, $p2 = \arg(P2)$. The phase difference can be corrected to

$$\begin{aligned} \alpha &= p + \frac{p1}{2} + \frac{p2}{3}, \{0 \le p \le \frac{\pi}{3}\} \\ &= p + \frac{p1}{2} + \frac{p2 \pm 2\pi}{3}, \{\pm\frac{\pi}{3} \le p \le \pm\frac{\pi}{2}\} \\ &= p + \frac{p1 \pm 2\pi}{2} + \frac{p2 \pm 2\pi}{3}, \{\pm\frac{\pi}{2} \le p \le \pm\frac{2\pi}{3}\} \\ &= p + \frac{p1 \pm 2\pi}{2} + \frac{p2 \pm 4\pi}{3}, \{\pm\frac{2\pi}{3} \le p \le \pm\pi\} \end{aligned} \tag{9}$$

Use this α for estimation of $\bar{\Delta f}$. Now repeat the Integer frequency offset estimation as discussed previously. This method improves the performance of Frequency offset estimation for low SNR cases.

III. Simulation Results

MATLAB simulations have been carried out by considering an OFDM system with 256 point FFT and CP length as $\frac{N}{4}$. Doppler Shift is considered to be Normalized Frequency Shift which is from 0.01 to 0.9.

A. Timing Synchronization Algorithm Results

For the basic Shmidl Cox algorithm at different SNRs, simulations have produced results in a system of 0 dB SNR AWGN noise which are only 70% correct. Efficiency is calculated as shown in table I.

TABLE I

Efficiency of Schmidl Cox at different SNRs					
SNR	12 dB	9 dB	6 dB	3 dB	0 dB
Efficiency	98	91	87	80	71

1) Comparison of Different Algorithms: After applying Multiple Correlations and Double Schmidl Cox methods, at 0 dB SNR AWGN noise, the system is 90% efficient, i.e., the timing recovered lies in the CP range, irrespective of any amount of frequency shift. After carrying out the simulations, efficiency for different algorithms at 0 dB SNR for different frequency Shifts are shown in table II. At -3 dB SNR, the timing recovery with more than 86% efficiency is possible with the proposed methods.

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TABLE II

Comparing the efficiencies of the algorithms at several Frequency Shifts			
Frequency Shifts $\frac{\triangle f}{f_s}$	Schmidl Cox	Multiple Correlations	Double Schmidl Cox
0.01	72	82	90
0.05	69	81	91
0.1	76	80	87
0.3	74	83	92
0.4	72	81	90
0.5	70	82	92
0.6	71	80	92
0.7	72	82	92
0.8	75	77	91
0.9	71	79	90

B. Frequency Synchronization Algorithm Results

Fractional offset has significant error at high noise levels. To overcome this, we steered to the previous improvements made in the timing synchronization algorithms. The multiple correlations have been considered, and mean of all these values is taken. With this new fractional estimate, Integer estimate is determined by using basic Schmidl Cox method. For 0 dB SNR and -3 dB SNR cases the error percentages for different relative frequencies are shown in tables III and IV respectively.

TABLE III

Error Percentage for 0 dB SNR		
Relative Fractional Frequency Shift $\frac{\triangle f}{max \triangle f}$	Schmidl Cox	Multiple Correlations
0.1	9.07	1.06
0.2	9.43	1.76
0.3	9.57	1.51
0.4	9.41	2.07
0.5	9.36	1.94
0.6	9.76	1.91
0.7	9.62	2.01
0.8	9.6	1.78
0.9	9.34	1.84

TABLE IV

Error Percentage for -3 dB SNR		
Relative Fractional Frequency Shift $\frac{\triangle f}{max \triangle f}$	Schmidl Cox	Multiple Correlations
0.1	21.87	5.21
0.2	20.76	4.76
0.3	21.43	5.10
0.4	20.65	5.41
0.5	20.60	4.92
0.6	20.24	5.04
0.7	21.61	4.84
0.8	21.01	4.61
0.9	22.17	4.73

IV. Conclusion

The Proposed improvisation algorithms over Schmidl Cox can be applied in any OFDM systems. Only difference instead of sending two identical halves in Preamble, 4 repetitions were used. Here only AWGN with different SNRs is simulated and their results were tabulated for different normalized frequency shifts. The paper also discussed various simulation results along with performance comparisons in terms of error percentage at different SNR. Here main considerations are the number of repetitions for the preamble and averaging of their individual Schmidl Cox outputs. Then further applying Schmidl Cox for one more time on the output of averaging gives us the better results. By considering more than 4 repetitions, further SNR improvement is possible.

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